

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

Set K consists of all the positive integers that are less than or equal to **160**, where none of the integers are repeated. If a number is selected at random from set K, what is the probability of selecting a number that is even and less than or equal to **50**?

Answer

- A. $\frac{5}{32}$
- B. $\frac{3}{16}$
- C. $\frac{5}{21}$
- D. $\frac{5}{16}$

Correct Answer: A

Rationale

Choice A is correct. It's given that set K consists of all the positive integers that are less than or equal to **160**, where none of the integers are repeated. Therefore, set K consists of the integers **1** through **160**, which is **160** numbers. Of these, the integers **1** through **50** are less than or equal to **50**. Therefore, there are **50** numbers that belong to set K that are less than or equal to **50**. Of these, half are even. Therefore, $\frac{50}{2}$, or **25**, numbers that belong to set K are even and less than or equal to **50**. The probability of selecting at random a number from set K that is even and less than or equal to **50** is equal to the number of numbers that belong to set K that are even and less than or equal to **50**, which is **25**, divided by the total number of numbers that belong to set K, which is **160**. It follows that the probability of selecting at random from set K a number that is even and less than or equal to **50** is $\frac{25}{160}$, or $\frac{5}{32}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the probability of selecting at random a number from set K that is less than or equal to **50**, not a number that is even and less than or equal to **50**.